NICHOLAS CARTER

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EDUCATION

Howard University | *B.S. in Computer Science, Minor in Japanese* **Washington, DC May 2026**

- Cum Laude, GPA: 3.5
- Study Abroad: J.F. Oberlin University, Tokyo, Japan

WORK EXPERIENCE

Citigroup | *Junior Generative AI Developer* **Sep 2026 - Present**

- Contribute to GenAI automation initiatives for AQUA, Citi's enterprise liquidity management platform supporting global treasury, fixed-income, and equities trading operations.
- Analyze liquidity and derivatives models and partner with business users to map end-to-end workflows, pinpointing manual bottlenecks in stress testing, funding analysis, and cost allocation.
- Scope and design GenAI solutions to automate manual derivatives processing and liquidity risk workflows, aiming to reduce processing time and manual intervention across trading businesses.

Research Institute for Tactical Autonomy (RITA UARC) | *Research Assistant* **Jan 2025 - May 2026**

- Developed perception and decision-making modules for autonomous UAV systems using Python, C++, and Linux, supporting mission-level autonomy across 100+ simulated flight scenarios.
- Analyzed 10,000+ sensor and telemetry data points to evaluate navigation, obstacle avoidance, and autonomy performance, identifying critical failure modes that informed system improvements.
- Built and optimized data-processing and performance-evaluation pipelines, reducing analysis turnaround time by 30%+ and enabling faster iteration of autonomy algorithms.
- Tested and validated autonomous behaviors in simulated and controlled environments, contributing to a 20%+ improvement in mission success and obstacle-avoidance reliability through iterative debugging and algorithm refinement.

Lockheed Martin | *Information Technology Intern* **May 2024 - Aug 2024**

- Implemented Documentation as Code practices by using GitLab Pages scraping to automate documentation updates, while leveraging Jira and Confluence to centralize agile documentation and strengthen product-team collaboration.
- Designed software delivery workflow diagrams and built an MkDocs site integrated with CI/CD, automating documentation publishing and improving process visibility.

REI Systems | *Quality Assurance Engineer* **Jun 2023 - Aug 2023**

- Developed and optimized an AI-powered resume matching system using Python and API integrations, improving candidate matching accuracy from approximately 62% to 91%.
- Built and maintained a database of 15,000+ candidate profiles, reducing manual screening effort and accelerating recruiter workflows.
- Implemented a binary reference method within a multi-label classification model using web-scraped data and documentation-as-code practices, mapping 25,000+ codes to descriptions to improve classification accuracy.

PROJECTS

Safe Space | *React, TypeScript, Supabase, AWS Amplify*

- Built a full-stack mental health platform supporting secure authentication and user progress tracking.
- Developed AI-powered journaling and mood analysis features that generated personalized coping strategies and behavioral recommendations.
- Implemented real-time dashboards, encrypted data storage, and notification systems to support user engagement and retention.

AI Japanese Learning Assistant | *React, TypeScript, Python*

- Built an AI-powered Japanese language learning platform with adaptive conversation practice and JLPT-style difficulty scaling.
- Implemented AI-driven error analysis and spaced repetition to personalize vocabulary and grammar practice.
- Designed a full-stack system with Python-based AI services to track learner progress and retention.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, JavaScript, TypeScript, SQL, HTML/CSS
- **AI/ML:** Generative AI, LLM APIs, Prompt Engineering, Multi-Label Classification, Data Pipelines
- **Frameworks & Cloud:** React, Supabase, AWS Amplify, REST APIs
- **Tools:** Git, GitLab, Linux, CI/CD, Jira, Confluence, SharePoint, MkDocs, Documentation as Code, Web Scraping
- **Spoken Languages:** English, Japanese

HONORS & AFFILIATIONS

- Jackie Robinson Foundation Scholar & Leadership Program (2022 - 2026)
- National Society of Black Engineers (NSBE), Member (2022 - Present)